MIL-STD-1215B
25 MAY 1988
SUPERSEDING
MIL-STD-1215A
30 MARCH 1973

MILITARY STANDARD

WATER PURIFICATION COMPOUNDS

AMSC N/A

FSC 6850

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Water Purification Compounds

MIL-STD-1215B

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, Attn: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

This standard is approved for use by all Departments and Agencies of the Department of Defense in the selection of items for application. It is intended to prevent the entry of unnecessary items (sizes, types, varieties) into the Department of Defense logistics system. This document is not intended to restrict any service in selecting new items resulting from state-of-the-art changes.

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1. SCOPE

1.1 Coverage. This standard is a presentation of nomenclature, formulas, physical and chemical properties, specification requirements, military and typical commercial uses, safety information, storage information and disposal information for water purification compounds. This standard does not include all of the items represented by the title or all those items which are commercially available. It does contain items preferred for use in the selection of water purification compounds for application by the Department of Defense.

1.2 Application. Water purification compounds are intended for military use for the purification of drinking water to render it free from disease bearing organisms.

1.3 Classification. The items in this standard are classified as water purification compounds on their ability to render water safe for drinking purposes.

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2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-C-114 - Calcium Hypochlorite, Technical
O-C-289 - Chlorination Kit, Water Purification (For Treatment of Drinking Water)
PPP-C-2020 - Chemicals, Liquid, Dry and Paste; Packaging Of

MILITARY

MIL-W-283 - Water Purification Tablet, Iodine
MIL-C-37873 - Chlorine Test Tablet, DPD Method, 100s
MIL-P-43327 - Paper, Book, Water Resistant (Text).

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation And Submission Of.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this standard to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

Title 29 - Department of Labor, Occupational Safety and Health Administration.
Title 40 - Protection of the Environment; Environmental Protection Agency.
Title 49 - Department Of Transportation; Hazardous Materials Regulations.

DOD 4145.19-R-1 - Storage And Materials Handling
DOD 4160.21-M - Defense Utilization And Disposal Manual
DOD 6050.5 - DOD Hazardous Materials Information System, Hazardous Item Listing.

TB MED 502 (DLAM 1000.2) - Occupational And Environmental Health Respiratory Protection Program.
TB MED 506 - Occupational And Environmental Health Occupational Vision.
TM 38-250 - Packaging, Materials Handling-Preparation Of Hazardous Materials For Military Air Shipment

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

Federal Insecticide, Fungicide, and Rodenticide Act (40 CFR 162)

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Reprints of certain regulations and application forms for registration of economic poisons may be obtained from the Registration Division, Environmental Protection Agency, Washington, DC 20460.)

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

Registry of Toxic Effects of Chemical Substances
Recommendation for Environmental Exposure Limits

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of DODISS specified in the solicitation.

AMERICAN CONFERENCE OF GOVERNMENT INDUSTRIAL HYGIENISTS (ACGIH)

TLVs® Threshold Limit Values for Chemical Substances in the Work Environment Adopted by American Conference of Government Industrial Hygienists (ACGIH) with Intended Changes.

Volume 13: Transactions of ACGIH Activities 1985 Conference.

(Application for copies should be addressed to American Conference of Governmental Industrial Hygienists, 6500 Glenway Avenue, Bldg D-7, Cincinnati, OH 45211.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Fire Codes

(Application for copies should be addressed to National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

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THE UNITED STATES PHARMACOPEIAL CONVENTION, INC./NATIONAL FORMULARY (USP/NF)

The United States Pharmacopeia/National Formulary

(Application for copies should be addressed to The United States Pharmacopeial Convention, Inc./National Formulary, Order Processing Department, P.O. Box 2248, 12601 Twinbrook Parkway, Rockville, MD 20852.)

(Nongovernment standards are generally available for reference from libraries. They are also distributed among nongovernment standards bodies and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

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3. DEFINITIONS

Not applicable.

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4. GENERAL REQUIREMENTS

4.1 Packaging data and labeling. All chemicals included in this standard shall be packaged in accordance with Federal Specification PPP-C-2020 and all applicable documents referenced therein. Shipping containers shall be labeled in accordance with current Department of Transportation (DOT) Hazardous Materials Regulations applicable to each chemical. When shipping by military aircraft the requirements of TM 38-250 shall apply. In addition, each item shall be packaged and labeled as specified in the applicable contract or order. All labels shall also comply with Hazard Communication Standard, 29 CFR 1910.1200 (f).

4.2 Hazardous materials information. DOD 6050.5, DOD Hazardous Materials Information System (HMIS) acquires, reviews, stores, and disseminates Material Safety Data Sheet (MSDS) information for all hazardous materials used by DOD. The contractual acquisition of a MSDS is accomplished through use of Federal Acquisition Regulation, paragraph 52.223-3, Hazardous Material Identification and Material Safety Data. The MSDS is prepared in accordance with the instructions in FED-STD-313; and shall comply with requirements of Hazard Communication Standard, 29 CFR 1910.1200 (g).

4.3 Safety.

4.3.1 Personal protective measures. The necessary respiratory, eye and skin protection to be used when handling chemicals shall be prescribed by the responsible installation industrial hygiene, medical and safety authorities.

4.3.1.1 Respiratory protection. Respirators, approved by the National Institute for Occupational Safety and Health (NIOSH) or the Mine Safety and Health Administration (MSHA) or by particular respiratory schedules of the Bureau of Mines (BM) for the compounds being used, may be employed for intermittent exposure or for supplementing other control measures (refer to TB MED 502 or DLAM 1000.2). Ventilation shall be adequate to remove hazardous concentrations.

4.3.1.2 Skin protection. Personnel using these compounds shall be provided with and required to use impervious gloves, sleeves, aprons, and boots whenever indicated. Protective creams and ointments commonly known as "barrier creams" may be of value in certain cases. However, barrier creams shall not be used to replace protective clothing. In case of contact with the skin, wash affected areas thoroughly with water. Eye lavages and emergency showers shall be located where there is a potential for direct contact with harmful chemicals.

4.3.1.3 Face and eye protection. Personnel using these compounds shall be provided with and required to wear chemical splash-proof safety goggles. In addition, face shields shall be provided and worn over the goggles if splashing could occur. In case of contact with the eyes, immediately irrigate with copious amounts of water for at least 20-30 minutes, and obtain medical attention. (Refer to TB MED 506.)

4.3.1.4 Training. Employers shall provide employees with training and information including MSDSs on all chemical items in their work area, in accordance with 29 CFR 1910.1200 (h), to ensure that employees know potential hazards of the chemicals with which they come in contact and the symptoms of

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exposure as well as how these chemicals affect the body and bodily functions. Employees shall be adequately trained to render first aid.

4.3.1.5 Exercises. Participation in exercises shall be stressed to demonstrate skills in the use of personal protective equipment and emergency response equipment.

4.3.2 Storage conditions. DOD 4145.19-R-1 describes general storage practices and requirements for hazardous materials in the DOD supply system. Specific requirements provided in the following paragraphs are supplementary in nature and shall be observed in consonance with the DOD storage regulations.

4.3.2.1 Flammable, combustible, pyrophoric and ignitable materials. A flammable material is generally any solid, liquid, vapor or gas that ignites easily and burns rapidly. Combustible materials are generally those that are difficult to ignite and burn slowly. The DOT, in Part 173, Subpart D, Section 173.115 of 49 CFR, defines a flammable liquid as one having a closed cup flash point below 100°F (37.8°C). A combustible liquid is defined, by DOT in the above reference, as one having a closed cup flash point at or above 100°F and below 200°F (93.3°C). A pyrophoric liquid is defined, by DOT in the above reference, as one that ignites spontaneously in dry or moist air at or below 130°F (54.5°C). Materials with flash points of 200°F or higher are to be considered as burnable. The Environmental Protection Agency (EPA), in Part 261, Subpart C, Section 261.21 of 40 CFR*, designates the criteria for flammable and combustible materials and oxidizers that exhibit the characteristic of ignitability (I). Liquids with closed cup flash points of less than 140°F (60°C) are defined by EPA as ignitable. The autoignition point (temperature) of a substance is generally defined as the minimum temperature required to initiate or cause self-sustained combustion in the absence of a spark or flame. Materials that ignite easily under normal industrial conditions are considered to be dangerous fire hazards. Such materials shall be stored in a manner to prevent ignition and combustion. Easily ignitable substances, such as reducing agents, shall be kept away from strong oxidizing agents. All containers shall be tightly sealed. It is important to provide adequate ventilation in storage areas, and to locate the storage areas of these items away from fire hazards. Ample fire-control equipment shall be easily accessible. Storage buildings, rooms and cabinets shall comply with provisions of the National Fire Codes. The building shall be electrically grounded and signs posted to prevent the lighting of matches or smoking in the area. Flammable storage areas shall be equipped with smoke or fire detection equipment.

4.3.2.2 Water-sensitive fire and explosive hazardous materials. These are materials that react on contact with water or steam to ignite or evolve heat or explosive gases. Such materials exhibit the characteristic of reactivity (R) as designated by the EPA in Section 261.23 of the above reference.* These materials shall be stored in well-ventilated, cool, dry areas. All containers shall be tightly sealed. These materials are a fire hazard in contact with water or moisture; therefore, it is essential that no sprinkler be used. Otherwise, the building shall conform to that required for storage of flammable materials. The building shall be waterproof, located on high ground, and separated from other storage areas.

*Refers only to materials that have become waste materials.

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4.3.2.3 Incompatible materials. Materials that are chemically incompatible shall be segregated in the storage of both serviceable and unserviceable items. The degree of segregation will depend upon DOD 4145.19-R-1 and local supplementary requirements that insure safe storage conditions. Hazardous storage compatibility codes are provided in the HMIS referred to in 4.2.

4.3.3 Chemical hazardous exposure limits. Chemical hazardous exposure limits for airborne concentrations of substances are obtained from the current TLVs® Threshold Limit Values for Chemical Substances in the Work Environment, adopted by the American Conference of Government Industrial Hygienists (ACGIH); current Occupational Safety and Health Administration (OSHA) Permissible Exposure Limits (PEL), 29 CFR, Part 1910, Subpart Z; and NIOSH Recommendation for Environmental Exposure Limits. Such information is also shown in MSDSs and the HMIS referred to in 4.2. The identity of sources establishing if a chemical is a carcinogen or potential carcinogen, for hazard communication purposes, is shown in 29 CFR 1910.1200 (d)(4).

4.3.4 Toxicity. Toxicity information for chemical compounds is available from various publications and from MSDSs, which are collected in DOD 6050.5 Hazardous Materials Information System.

4.3.4.1 EPA Toxic (T). Some chemical compounds have been designated by the EPA as toxic (T) in accordance with the criteria shown in Part 261, Subpart B, Section 261.11(a)(3) of 40 CFR.* Some commercial chemical products are listed as toxic under Subpart D, Section 261.33(f).

4.3.4.2 EPA Acute Hazardous Toxicity (H). Some chemical compounds have been designated by the EPA as acute hazardous (H) in toxicity in accordance with the criteria shown in Subpart B, Section 261.11(a)(2) of 40 CFR.* Some commercial chemical products are listed as acute hazardous in toxicity under Subpart D, Section 261.33(e).

4.3.4.3 EPA Extraction Procedure Toxicity (EP Toxicity) (E). Some chemical contaminants, that are not listed in Subpart D, have been designated by the EPA as exhibiting the characteristic of EP toxicity, and are listed in Section 261.24 Table 1 of 40 CFR.*

4.3.4.4 Hazardous toxic constituents. A list of chemical compounds and substances, shown to have toxic effects on humans or other life forms, is contained in Appendix VIII to 40 CFR Part 261; and the Registry of Toxic Effects of Chemical Substances.

4.4 Pollution and disposal.

4.4.1 Pollution potential. All items described in this standard shall be assumed to have a pollution potential. However, to minimize this potential, the proper use, storage and disposal methods shall be strictly followed.

4.4.2 Disposal of excess or unserviceable material. To minimize disposal problems, it is recommended that no more than a one year's supply of each item listed in this standard be stocked. When stocks have been declared excess or unserviceable, they will be disposed of in accordance with the Defense Utiliza-

*Refers only to materials that have become waste materials.

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tion and Disposal Manual, DOD 4160.21-M, and applicable DOD Policy Memoranda. Guidance can be obtained from your servicing Defense Reutilization and Marketing Office (DRMO) on procedures required for proper reporting and turn-in.

4.4.3 Disposal and storage of hazardous wastes. Items are classified and managed as hazardous wastes as defined by the Resource Conservation and Recovery Act (RCRA) (Public Law 94-580). Items have been identified as meeting the characteristics (i.e., ignitable, corrosive, reactive or EP toxic) or are listed (i.e., toxic or acute hazardous) according to Identification and Listing of Hazardous Waste, Part 261; 40 CFR; or have been determined to be hazardous wastes by declaration of the Defense Reutilization and Marketing Service (DRMS) in accordance with procedures set forth in DOD 4160.21-M. Disposal of such items shall be managed in accordance with the Installation Environmental Office, the DRMO, or the Safety and Health Office to insure proper reporting of disposal and treatment actions to the US EPA and State; and shall be managed in accordance with Federal, State and local laws. The three main disposal methods are turn-in to the DRMO, on-post disposal by installation personnel, or disposal by commercial contract. Hazardous wastes that cannot be used, or disposed of as stated in 4.4.3.2, shall be stored under environmentally safe conditions until suitable methods of disposal are determined. Short-term storage (less than 90 days) requires proper containment (i.e., packaging and facilities) in accordance with Section 262.34, Part 262 of the above reference. Long-term storage (greater than 90 days) requires permitting by the EPA or by the state under Public Law 94-580 (RCRA), in compliance with the requirements of 40 CFR Parts 264 and 265. Physical custody will be accomplished by the activity with conforming storage or most nearly conforming storage. When physical custody is in question, the Post Commander will make the final decision. In all cases where the wastes are to be collected, stored, transported and disposed of at a state or local permitted disposal facility, the identity and description of the waste shall be maintained and recorded in accordance with Part 262 of the above reference. Transportation of the waste shall be in accordance with Part 263 of the above reference, Standards Applicable to Transporters of Hazardous Waste.

4.4.3.1 Cleanup of liquid spills. To control the migration of spilled or leaking liquids, dike around the item with an inert, dry absorbent (e.g., clay, sawdust or vermiculite) or follow installations spill plans (Spill Prevention Control and Countermeasure Plan and Installations Spill Contigency Plan). Control entry to the spill site and segregate salvageable materials away from the spill area. Initiate waste cleanup operations immediately in accordance with local procedures. The residue shall be safely handled and transported to an approved or permitted disposal or storage facility. Packaging, labeling, transportation and record-keeping requirements for this waste material are determined by the appropriate Federal and State agencies and local procedures. It is recommended that all activities involving disposal preparation and transportation to commercial facilities be properly coordinated with the appropriate Federal and State agencies responsible for health and environmental aspects of hazardous materials. It is imperative that the proper description of waste accompany the packaged item at all times. Final disposal of the waste item shall be accomplished by reutilization, transfer, donation or sales by DRMS in accordance with DOD 4160.21-M or by ultimate disposal as described in 4.4.3.2. Spill residue, including contaminants, to be turned in to the DRMO shall first

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be properly identified, containerized, and labeled. For large scale spills that grossly contaminate the environment, the Chemical Transportation Emergency Center (CHEMTREC), can be called for assistance. Applicable procedures of the local spill control plan shall be followed. Necessary respiratory, eye, and skin protection measures are to be used while performing cleanup operations.

4.4.3.2 Ultimate disposal. Ultimate disposal shall be accomplished at a permitted or approved hazardous waste treatment or disposal facility designated by the Installation Environmental Office, DRMO, or Safety and Health Offices.

4.4.4 DISCLAIMER. RECOMMENDED DISPOSAL INSTRUCTIONS ARE FORMULATED FOR USE BY ELEMENTS OF THE DEPARTMENT OF DEFENSE. THE UNITED STATES OF AMERICA IN NO MANNER WHATSOEVER EITHER EXPLICITLY OR IMPLICITLY WARRANTS, STATES, OR INTENDS SAID INSTRUCTION, TO HAVE ANY APPLICATION, USE OR VIABILITY BY OR TO ANY PERSON OR PERSONS CONTRACTING OUTSIDE THE DEPARTMENT OF DEFENSE OR ANY PERSON OR PERSONS CONTRACTING WITH ANY INSTRUMENTALITY OF THE UNITED STATES OF AMERICA AND DISCLAIMS ALL LIABILITY FOR SUCH USE. ANY PERSON USING THESE INSTRUCTIONS WHO IS NOT A MILITARY OR CIVILIAN EMPLOYEE OF THE UNITED STATES OF AMERICA SHOULD SEEK COMPETENT PROFESSIONAL ADVICE TO VERIFY AND ASSUME RESPONSIBILITY FOR THE SUITABILITY OF THESE INSTRUCTIONS TO THEIR PARTICULAR SITUATION REGARDLESS OF SIMILARITY TO A CORRESPONDING DEPARTMENT OF DEFENSE OR OTHER GOVERNMENT SITUATION.

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5. DETAILED REQUIREMENTS

5.1 Name. Water Purification Tablet, Iodine

5.1.1 Technical description. Iodine water purification tablets are a double convex type having the following ingredients: tetraglycine hydroperiodide, sodium acid pyrophosphate, anhydrous, and talc.

5.1.2 Specification. Military, MIL-W-283, Water Purification Tablet, Iodine.

5.1.2.1 Requirements. The military specification requirements for water purification tablet, iodine is shown in Table I and II.

TABLE I. Tetraglycine Hydroperiodide - requirements.

Ingredient	% by wt	
	Min	Max
Active iodine	39.50	42.60
Ash	---	0.50
Total nitrogen	7.35	7.85
Titratable iodine after 7 days storage in a tightly sealed bottle at 130°F	34.50	---

5.1.2.1.1 Sodium acid pyrophosphate, anhydrous. Sodium acid pyrophosphate, anhydrous ($Na_2H_2P_2O_7$) shall meet the requirements of the Food Chemical Codex grade.

5.1.2.1.2 Talc. The inert material shall consist of non-asbestiform talc, United States Pharmacopeia (USP) suitable for its intended use. It shall be selected to provide good tableting properties without affecting the solubility, friability and stability of the tablets beyond the specified limits. In addition, the talc shall lose not more than 0.3 percent of its weight when tested as specified.

5.1.2.1.3 Form. The tablets shall be uncoated, be a firm double convex form and shall contain no added coloring material.

TABLE II. Water Purification Tablet, Iodine - composition requirements.

Ingredient	Milligrams	
	Min	Max
Tetraglycine hydroperiodide	19.3	21.3
Sodium acid pyrophosphate, anhydrous	82.5	92.3
Talc, USP	---	6.0

5.1.2.2 Physical and chemical requirements.

5.1.2.2.1 Weight. The average tablet weight shall not exceed 120 milligrams when tested as specified. The individual tablet weight deviations shall conform to the tolerances for tablet weight as outlined in the Weight Variation Test for Tablets referenced in the USP.

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5.1.2.2.1 Loss on drying. Tablets shall have a maximum loss on drying of 0.20 percent when tested as specified, regardless of the maximum allowable loss on drying of the individual components of the tablet.

5.1.2.2.2 Titratable iodine. Tablets shall yield not less than 7.60 milligrams nor more than 9.00 milligrams per tablet of titratable iodine when tested as specified.

5.1.2.2.3 Solubility. Tablets, when tested as specified shall dissolve in not more than 200 seconds. If the tablets do not dissolve within the specified time, they will be acceptable providing the titratable iodine in solution at the end of 200 seconds is not less than 7.60 milligrams when tested as specified.

5.1.2.2.4 Friability. The tablets shall not lose more than 10.0 percent in weight when tested as specified.

5.1.2.2.5 Stability.

5.1.2.2.5.1 Loss of iodine after storage. Tablets as packaged in the sealed, glass bottle after 7 days storage at 130°F shall show a loss of not more than 5.0 percent of the original titratable iodine when tested as specified.

5.1.2.2.5.2 Loss of solubility after storage. Tablets as packaged in the sealed, glass bottle after 7 days storage at 130°F, when tested as specified, shall dissolve in not more than 300 seconds. If the tablets do not dissolve within the 300 second time, they will be acceptable providing the titratable iodine in the 300 mL aliquot at the end of 300 seconds is not less than 95.0 percent of the original titratable iodine.

5.1.3 Use. Iodine water purification tablets are intended for military use for the purification of water in canteens.

5.1.4 Safety. General safety and hygienic measures shall be exercised in the use of water purification tablets, iodine.

5.1.5 Storage. Water purification tablets, iodine, shall be stored in a cool, dry place in tightly closed containers away from sources of heat.

5.1.6 Disposal. For appropriate procedure, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

5.2 Name. Chlorination Kit, Water Purification
(For Treatment of Drinking Water)

5.2.1 Technical description. Chlorination kit, water purification, (for treatment of drinking water) shall consist of: tubes of calcium hypochlorite, chlorine test tablets, DPD No. 1, color comparator and instruction sheet.

5.2.2 Specification. Federal, O-C-289, Chlorination Kit, Water Purification, (For Treatment of Drinking Water).

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5.2.2.1 Requirements. The federal specification requirements for chlorination kit, water purification, (for treatment of drinking water) is shown in Table III.

TABLE III. Chlorination Kit, Water Purification (For Treatment of Drinking Water) - requirements.

Component	Quantity
Tubes of calcium hypochlorite	100
Chlorine test tablets, DPD No. 1	150
Color comparator	1
Instruction sheet	1

5.2.2.1.1 Tubes of calcium hypochlorite. The tubes specified in Table III shall be filled with calcium hypochlorite such that each tube contains 0.60 ± 0.05 grams of calcium hypochlorite when tested as specified.

5.2.2.1.2 Calcium hypochlorite. The calcium hypochlorite shall conform to the requirements of federal specification, O-C-114, Calcium Hypochlorite, Technical. The calcium hypochlorite shall be registered with the Environmental Protection Agency in accordance with the Federal Insecticide Fungicide and Rodenticide Act.

5.2.2.1.3 Tubes. The tubes shall be made from clear glass tubing having an outside diameter of 8.55 to 8.95 mm and a wall thickness of 0.8 ± 0.05 mm. (Note: Using the above tubing, a theoretical minimum internal diameter of 6.85 mm and a maximum internal diameter of 7.45 mm with an average internal diameter of 7.15 mm is obtained). The tubes shall be approximately 70 mm long and shall be hermetically sealed by fusion of the glass with no leaks when tested as specified. The tubes shall have near the center, a transverse file mark extending 360 degrees around the tube and of sufficient depth to permit the tube to be broken with a clear right-angle break without splintering or shattering. Each tube shall be marked with the following information:

Calcium hypochlorite

Contents of tube normally sufficient to provide
2 mg/L dosage for one sterilizing Lyster bag

30-35 gallons of water
(Date packaged)

5.2.2.1.4 Chlorine test tablets, DPD No. 1. The chlorine test tablets, DPD No. 1, quick dissolving shall conform to the requirements of MIL-C-37873 except that each tablet shall be hermetically sealed in a foil-polymer laminate. (The sealed package may be in strips not exceeding 10 tablets per strip.)

5.2.2.1.5 Color comparator. The color comparator shall be formed of colorless plastic and shall hold a 10 mL water sample without leaking. It shall have six stepped sections or cells, each with a viewing window and marked in black: 1, 1.5, 2, 3, 5 and 10 respectively. Each cell window shall present a minimum viewing area of 100 mm^2 and shall be bordered in black. Alongside the cell windows shall be a similar row of color standard windows. The color of

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the standard shall match the color seen through each window when a water sample contained that amount of available chlorine is poured into the comparator and a DPD tablet added, when tested as specified.

5.2.2.1.6 Instruction sheet. The instructions shall be legibly printed with black ink on white water resistant paper conforming to MIL-P-43327, Paper, Book, Water Resistant (Text).

5.2.3 Use. Chlorination kit, water purification is intended for use in the disinfection of water in Lyster bags and small containers by chlorine in order to render water safe for drinking purposes.

5.2.4 Safety. General safety and hygienic measures shall be exercised in the use of chlorination kit, water purification.

5.2.5 Storage. Chlorination kit, water purification shall be stored in a cool, dry place away from sources of heat. Decomposition of calcium hypochlorite is accelerated at temperature above 90°F.

5.2.6 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices.

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6. NOTES

6.1 Intended use. This standard is intended to cite nomenclature, formulas, physical and chemical properties, specification requirements, military and typical commercial uses, safety information, storage information and disposal information for Water Purification Compounds preferred for application by the Department of Defense.

6.2 Subject term (key word) listing.

Calcium hypochlorite, technical
Chlorination kit, water purification, (for treatment of drinking water)
Chlorine test tablet, DPD method, 100's
Color comparator
Exposure limits, hazardous chemicals
Hazardous wastes, disposal and storage of
Information, hazardous chemicals
Iodine water purification tablet
Safety, hazardous chemicals
Sodium acid pyrophosphate, anhydrous
Talc
Tetraglycine hydroperiodide
Water purification chlorination kit (for treatment of drinking water)
Water purification tablet, iodine

6.3 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 Abbreviations. The use of abbreviations shall be in accordance with MIL-STD-12 where applicable. Metric system abbreviations and symbols shall be in accordance with ASTM E 380.

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Preparing activity: Army - EA Project Number 6850-1029

Custodians:
Army - EA
Navy - YD
Air Force - 68

Review activities:
Army - ME, GL, MD
Navy - SH
DLA - GS

User activities:
Army - AR, CE, CR, SM
Navy - MC, MS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1215B	WATER PURIFICATION COMPOUNDS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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